One More Pivot

Farsighted Search When Trials Are Scarce

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Abstract

Firms search for strategy the way Callander's searchers probe an unknown correlated landscape: each trial is a strategy actually lived, and the last one is the one committed to. Field experiments show the search is short — roughly three quarters of startups pivot zero times in a year, a fifth exactly once, and six to eleven percent more than once — so the empirically relevant problem is not an infinite horizon truncated but a game of at most two moves, and that game has recently been solved exactly on the canonical mean-reverting landscape. This paper translates the solution into operating rules: when a struggling incumbent should pivot far, how wide to bracket before the final commitment, the explicit performance ceiling above which the grass is provably not greener, and the threshold — strictly below the performance of positions already held — at which a disappointing pivot should be abandoned rather than repaired. The rules are graded and falsifiable, the interior option is worth about one percent of expected value at its best, and the same field experiments that motivate the model rhyme with its central prediction: better search produces exactly one pivot, not many.

1 Callander's problem

Callander (2011b) posed the problem this way. Strategies sit on a line, and the mapping from strategy to outcome is the realized path of a Brownian motion: known where it has been tried, unknown elsewhere, with nearby strategies performing similarly. A searcher tries a strategy, lives its outcome, and decides whether to try another. Trials are not laboratory queries; each is a policy enacted, a product shipped, a position occupied. In the companion political-economy version (Callander, 2011a), the same machinery explains why policymaking is incremental and why search stalls at mediocre policies that nobody can improve on locally.

The workhorse of the tradition that grew from these papers is myopic: each period's move maximizes the current outcome, and the theory asks where such a searcher drifts, when she stops, and what she leaves undiscovered. The survey of Bardhi and Callander (2026) lists farsighted search — a searcher who plans her few remaining trials jointly — among the tradition's open methodological problems, and recent progress has bought tractability with restrictions: contiguous movement along a vein (Urgun and Yariv, 2025; Wong, 2025), or exact few-point placement under an information objective (Bardhi, 2024). The management literature on rugged landscapes (Baumann et al., 2019) runs on simulation for the same reason.

2 The evidence: search is short

Field evidence indicates that real ventures make very few strategic pivots. In the randomized trial of Camuffo et al. (2020), 74% of firms pivoted zero times over roughly a year, 19% exactly once, and 7% more than once. The four-trial replication (Camuffo et al., 2024), with 759 firms, found the same shape: mode zero, 11% more than once, 6% more than twice. Practitioner folklore agrees on the mechanism — "the true measure of runway is how many pivots a startup has left" (Ries, 2011) — though no peer-reviewed cost-per-pivot measurement exists, and pivot counts should be read with the construct caution of Kirtley and O'Mahony (2023), who show pivots are often accumulated increments rather than single jumps.

The treatment effect is the striking part. Teaching firms a scientific approach to search raised the probability of pivoting exactly once by 8.3 percentage points, left twice unchanged, and *lowered* the probability of pivoting more than twice by 3.7 points (Camuffo et al., 2024). Better search does not produce more search. It produces one good pivot.

A searcher holding an incumbent position, one further trial, and a final commitment — at most two pivots — is therefore not a truncation of the empirically relevant problem. It is the empirically relevant problem.

3 The model, in business terms

The landscape is the exponential of a stationary Ornstein–Uhlenbeck path: log-performance mean-reverts, so there is no free lunch far from what is known, and the exponential makes rewards right-skewed, as venture outcomes are. The firm holds an incumbent position of known performance, may run one more trial anywhere, and then must commit — to the incumbent, to the tested alternative, or to an untested position between or beyond them. The payoff is the performance of the final position, not the best ever visited: the strategy committed to is the one lived with.

Mean reversion is what makes the pivot a trade. On a pure random walk landscape, moving far from the incumbent keeps its expected performance while adding variance without bound, and a skew-loving searcher would walk to the horizon; the model would recommend infinite pivots and contradict the evidence in the first sentence. Under mean reversion, distance buys variance but surrenders expected performance toward the population median, and the exact solution below prices that trade.

The exact solution is developed in a companion paper (Cotton, 2026); here we state what it says operationally.

4 Operating rules from the exact solution

Go far on bad news. With one trial and then commitment, the optimal placement has three phases. An incumbent below the median performance should be abandoned with a distant trial: its information is not worth anchoring on. A strong incumbent, above an explicit threshold, should be held without experimentation. In between, the trial goes a distance set by the mean-reversion scale of the industry, not by the firm's ambition.

Bracket before committing. With two trials, the searcher first establishes a bracket of two known positions and then faces the signature question: commit to a tested position, or to untested ground between them? Every interior position matches the expected performance of an explicit tested-side alternative while carrying strictly more uncertainty, so the interior is preferred exactly when the payoff is convex — when upside is what is being bought. For a risk-averse operator judged on expected utility rather than expected value, the same identity flips the verdict: the grass between two tested positions is provably browner.

The ceiling. Above an explicit performance level, the grass is not greener anywhere: mean reversion caps the expected performance of every unvisited position, and no amount of variance-loving compensates. A firm whose incumbent clears the ceiling has nothing to gain from further search, and the ceiling is a formula, not a sentiment.

A bad pivot is abandoned, not repaired. Entering a bracket is not a commitment. If the trial between two known positions comes in at or below the median, every further move inside the spoiled bracket — including repairing the disappointing position — is weakly dominated by leaving for the best outside option; strictly, if the draw is strictly below median. The general rule is a threshold: stay inside exactly when the interior draw beats a cutoff that sits strictly *below* the performance of the positions already held (in a representative calibration, 0.63 against anchors of 0.70). A pivot that lands somewhat worse than both of its neighbors is still worth developing; one that lands badly is walked away from. The asymmetry is the model's sharpest managerial statement.

The size of the cleverness. The interior option — the right to commit to untested ground between two tested positions — is worth about one percent of expected value at its best. Most of the value of a second trial is the trial itself, not the sophistication of where the commitment lands afterward. The model prices its own subtlety, and prices it small.

5 The rhyme with the experiments

The exact solution has a revisit window: an interval of incumbent performance in which the optimal play is one trial and a considered commitment, bounded below by the flee region and above by the ceiling. The randomized trials found that better search shifts firms toward exactly one pivot and away from both zero and many. The two shapes rhyme. We state this as a rhyme and not an identity: the treatment effect in Camuffo et al. (2024) is a nonlinearity in an intervention's effect on pivot counts, not a causal estimate that a third pivot destroys value, and the model's window is an interval in incumbent quality, not in pivot count. A design that measured incumbent quality at baseline could test the correspondence directly.

6 One last trial in operations

The same three-move structure appears wherever a process owner holds an incumbent operating point, has time for one more pilot, and must then deploy: the final "shot" is the deployment decision, not a laboratory run. The rules above apply verbatim, with one qualification that decides applicability: the commitment set must genuinely include untested interpolated settings within an approved operating range. If deployment is restricted to individually validated settings, the interior option is off the table and the problem reduces to the simpler tested-side rules.

For the experimental-design community the model is a benchmark rather than a rival. The trial-placement step maximizes the expected value of the best deployable setting after one more observation, which is the knowledge-gradient criterion of Frazier et al. (2008) with an exponential objective; no new selection principle is claimed. What the model contributes is an unusually explicit instance — exact interior maximization, an exact phase diagram — against which knowledge-gradient implementations and their approximations can be checked, and possibly undercut in computation, since on a Markov landscape a new observation changes only its containing interval.

7 Closing

Neither the myopic workhorse nor the infinite-horizon limit describes the search the evidence shows firms conducting. The scarce-trial regime is its own theory, it now has an exactly solved case, and its rules are concrete: go far on bad news, bracket before committing, respect the ceiling, abandon bad pivots at a threshold below your anchors, and expect the cleverness premium to be small. We leave to future work the k-shot game, information-motivated trials, and the field test the rhyme of Section 5 invites.

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